

Best Subset for Predicting Public Undergraduate Institution In-State Tuition

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Read in Data

Read csv from this link into a dataframe `df`.

Code book available here.

Each row is unique by column UNITID.

1. Data Cleaning

For this analysis, we will focus on only Public school.

```
##           CONTROL  min  median   max      sd  n missing
## 1 Private for-profit 5160 16449.0 51400 7555.020 80      14
## 2 Private not-for-profit    0 38400.0 85000 16047.532 778      6
## 3           Public    910  7019.5 21989  3804.278 758      4
```

```
## Number of Public schools = 758
```

```
## Number of columns = 124
```

1st Cleaning

Next cleaning will does the following.

- filter the schools down to only Public
- turn any categorical variables into dummy variables (if there are n levels, there will be n-1 extra variables)
- remove the original categorical variables because we will use the created dummies for analysis from now on.

```
## Number of columns after 1st cleaning = 136
```

2nd Cleaning

Check NAs, we don't want to use variables that have a lot of missing rows (over 5%). Variables like Men, Women we will only keep Men. We will also remove other tuition data like out-of-state tuition, published tuition, and other uninterested variables.

```
## Number of columns after 2nd cleaning = 64
```

3rd Cleaning

We will now do a third time cleaning:

- Remove any non-numerical variables (to remove extra categorical variables we're not interested in. If the number of columns decrease, should check what extra columns got removed). In this case, we removed ATHASSOC (Member of National Athletic Association) because we have one for NCAA already, removed CARNEGIEIC (Institutional Classification) because we have program mix and this variable is over-complicated.
- Remove any row with NAs

```
## Number of rows after 3rd cleaning = 717
```

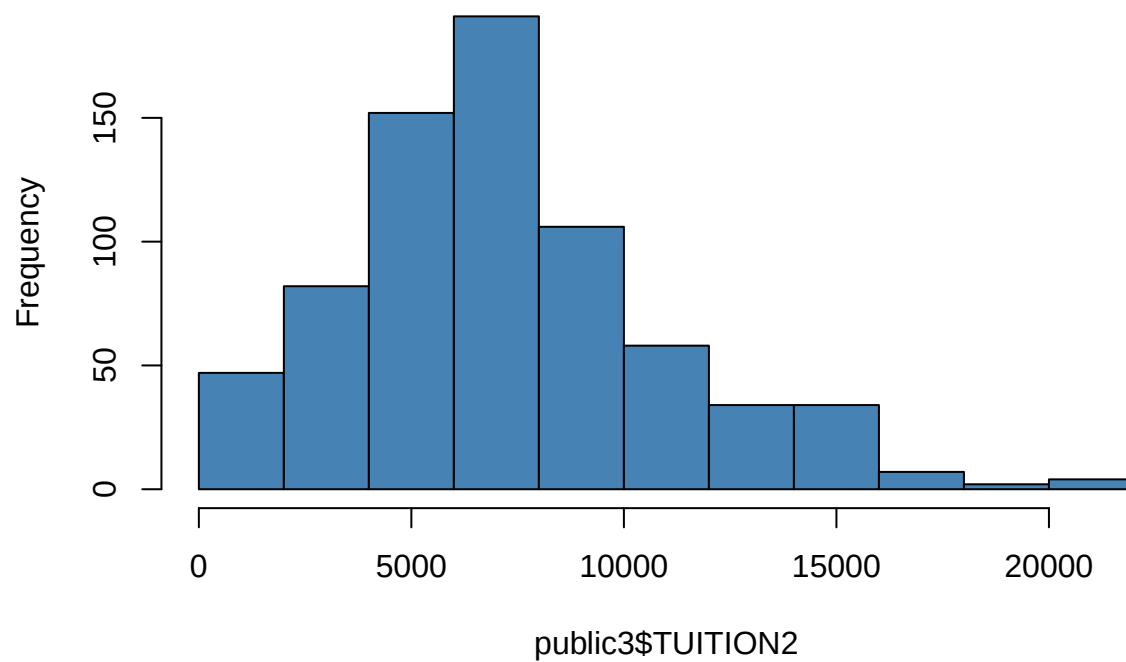
```
## Number of columns after 3rd cleaning = 62
```

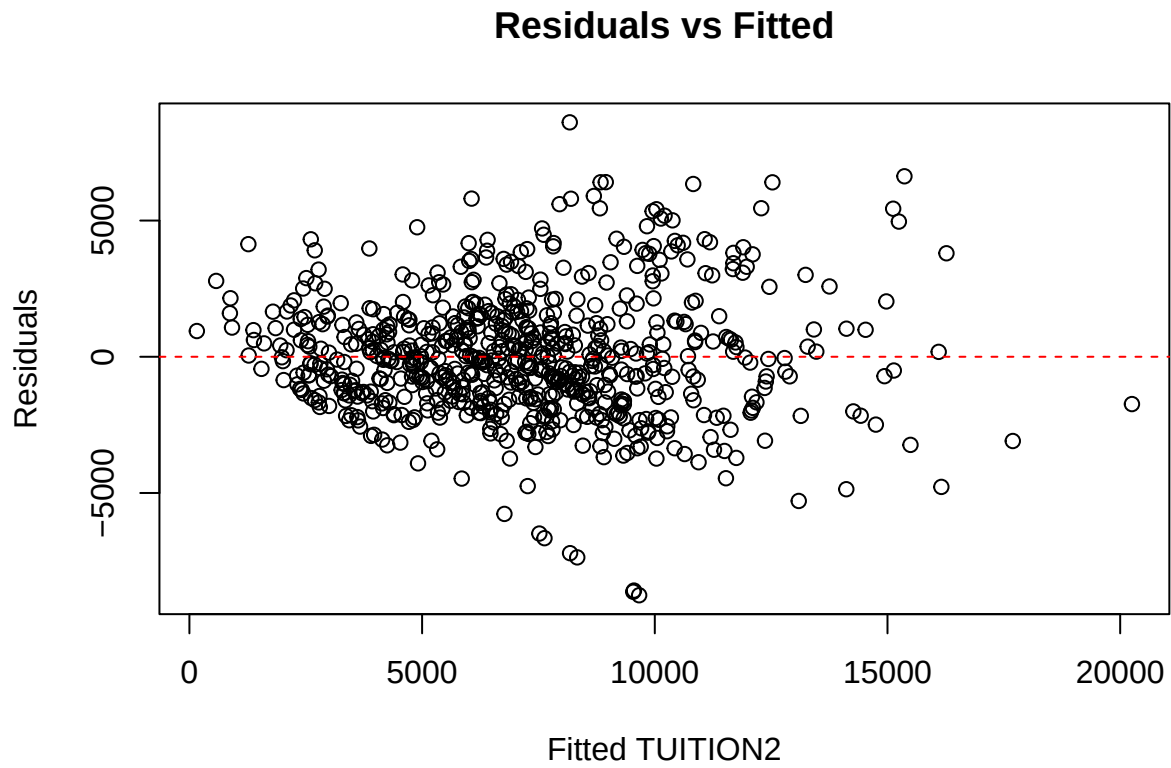
2. Data Transformation

Response Variable: TUITION2

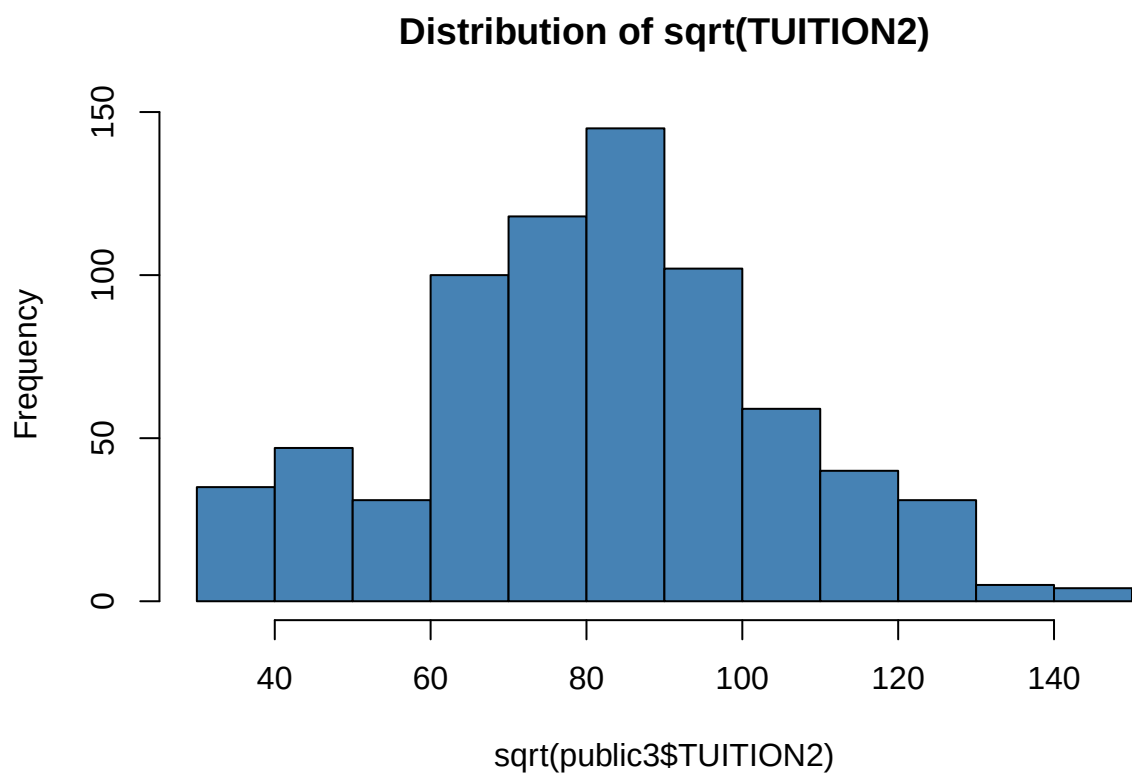
Tuition price is almost always right-skewed and usually need log-transformation but we will check just in case. Here's the distribution of the original TUITION2.

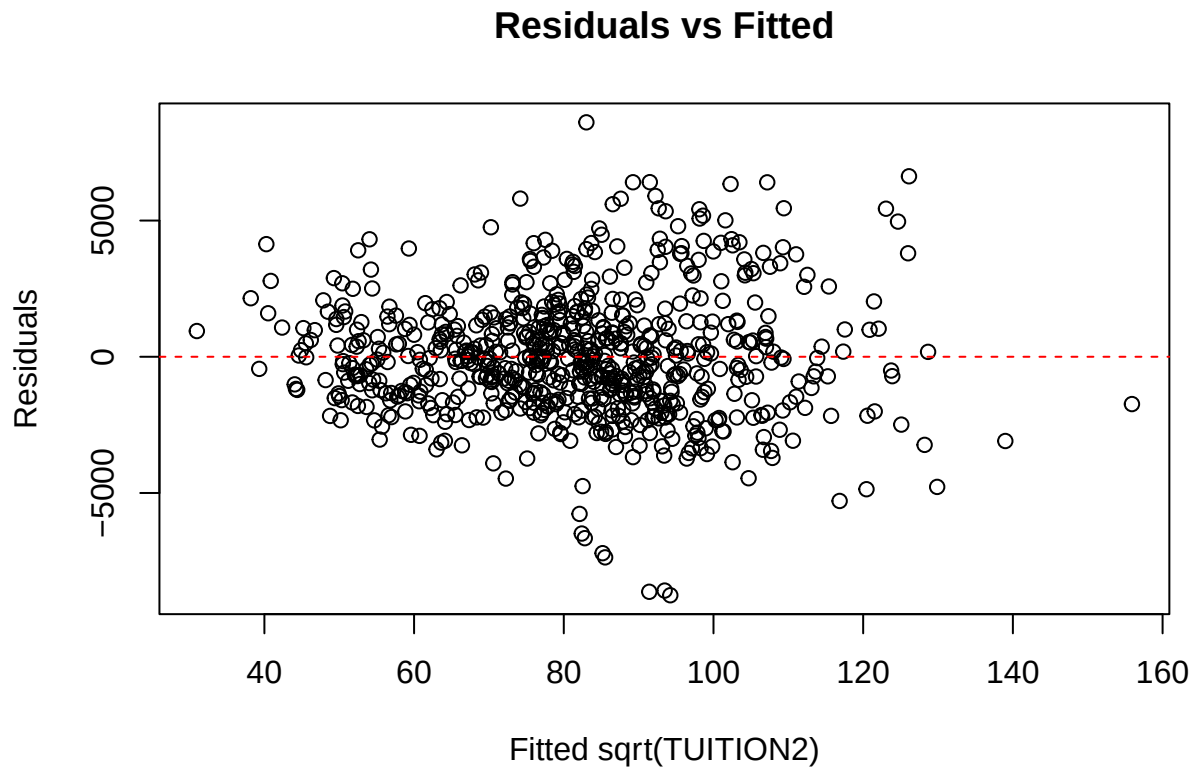
Distribution of TUITION2





The histogram looks fairly right-skewed, so we apply square-root transformation. Log-transformation would be too aggressive and flip the tail.



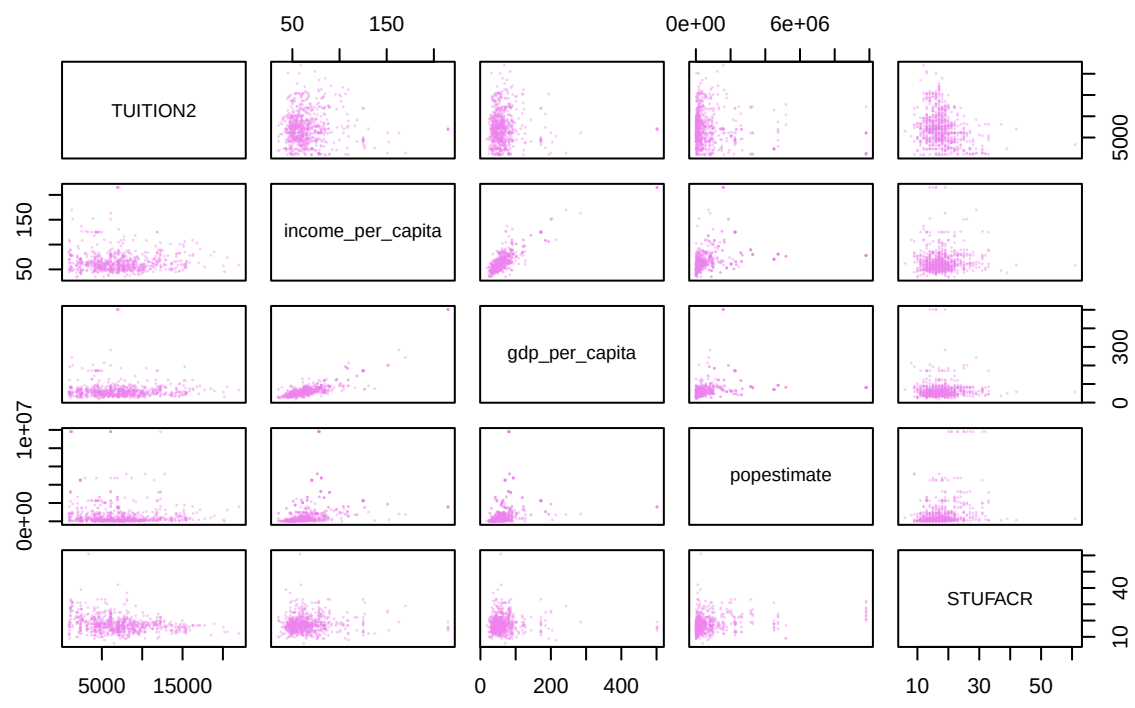


The residual plot with the **square-root TUITION2** looks fairly random with mild heteroskedasticity (variance increases with fitted values). The histogram is much more improved, looking more bell-shaped. Thus, we apply the square root transformation.

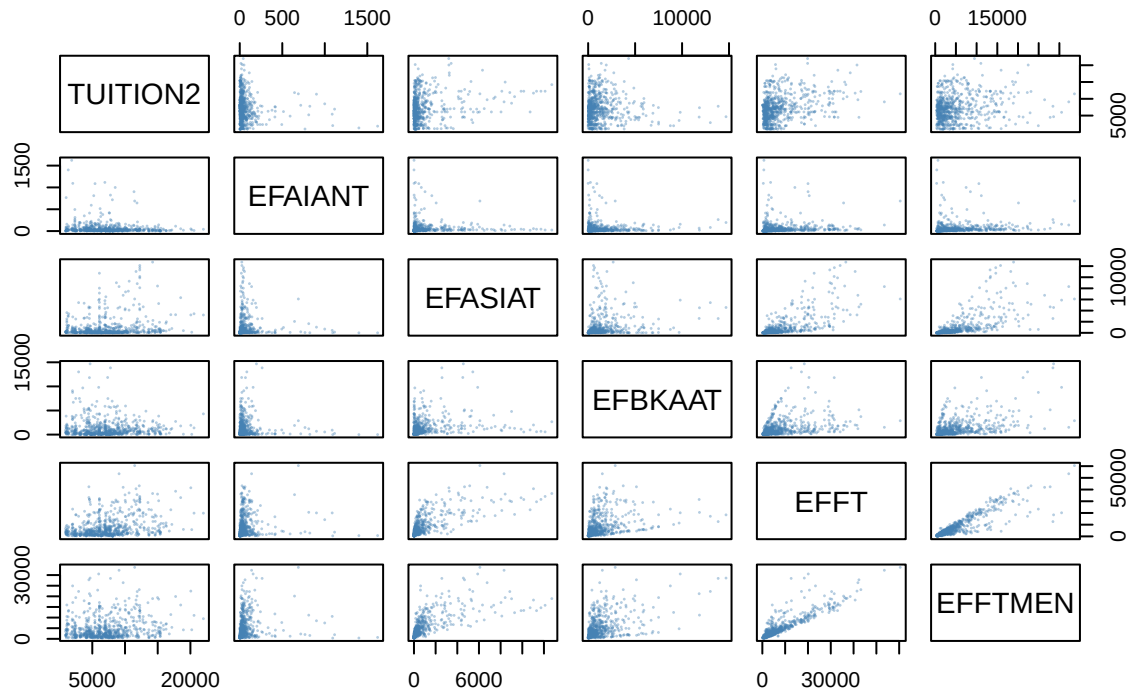
Numeric variables vs TUITION2

We will now check numerical variables in pair plots with TUITION2 to see if we need to perform transformations.

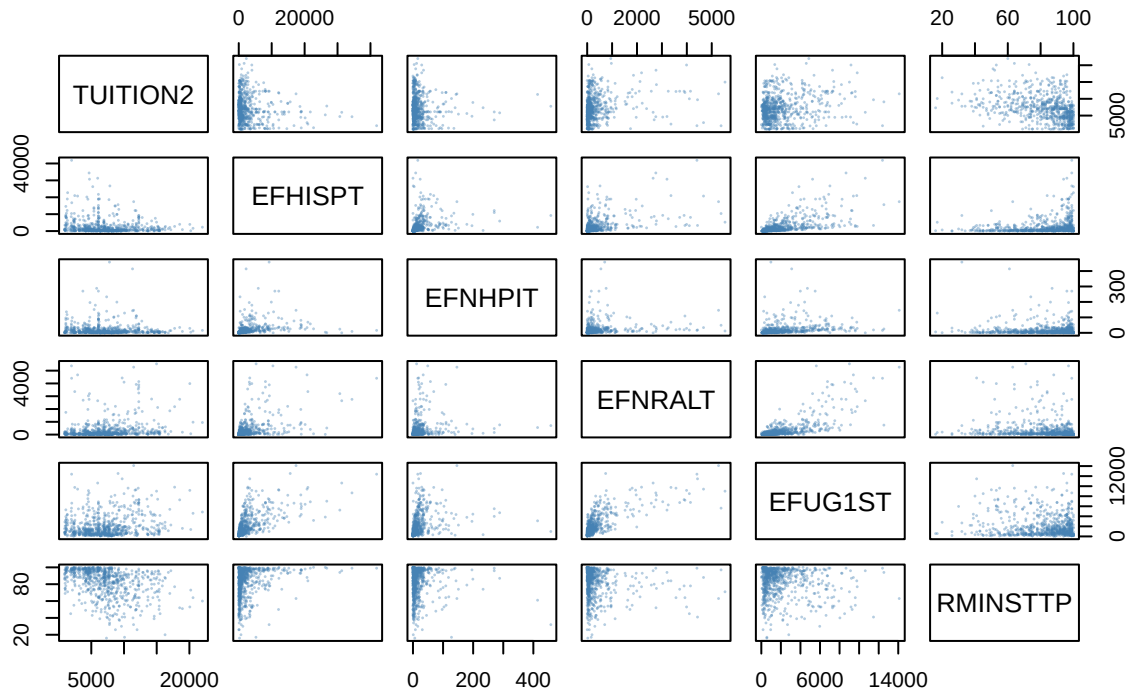
Matrix Plot: Institution Profile & Resources vs In-State Tuition



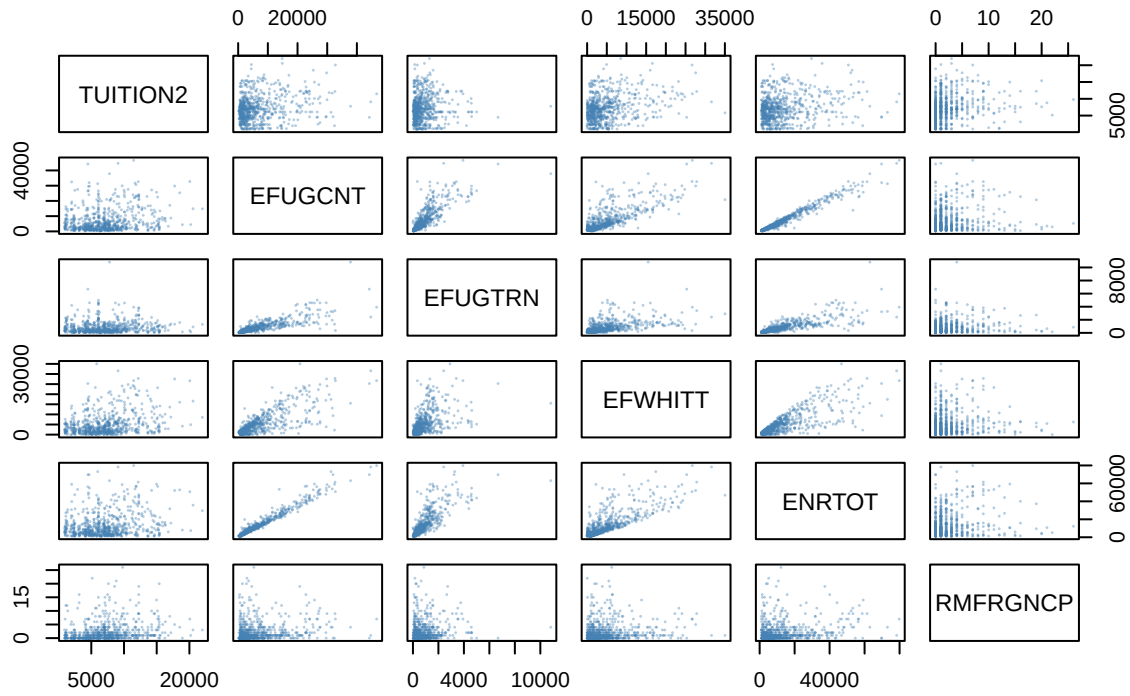
Matrix Plot: Enrollment Variables 1 vs In-State Tuition



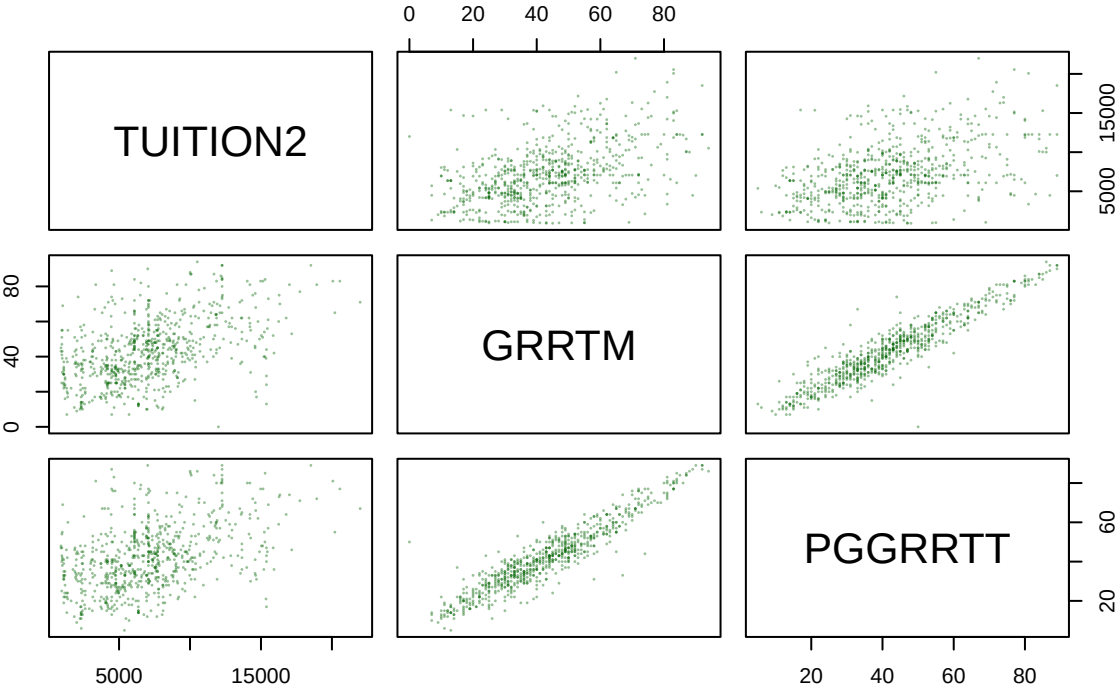
Matrix Plot: Enrollment Variables 2 vs In-State Tuition

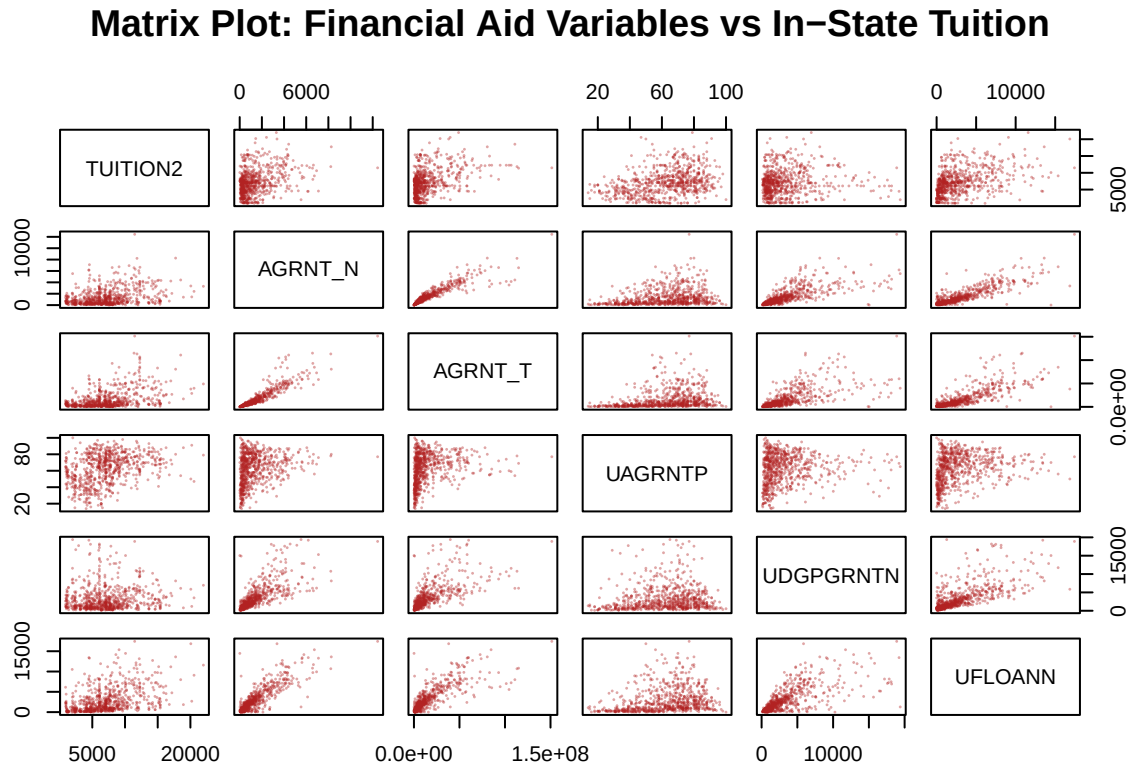


Matrix Plot: Enrollment Variables 3 vs In-State Tuition



Matrix Plot: Admissions & Outcomes vs In-State Tuition





Most of our numerical variables have extreme right skewness. Thus we will:

- Log-transform $\log(x + 1)$ on counts or total dollar amounts (Enrollment & demographic counts, grants, aid and economic resources) since they are heavily right-skewed
- Square-root transformation $\sqrt{x}$ on ratios like STUFACR since it has moderate skewness
- No transformation on variables like graduation rates (GBA4RTT, GBA6RTT, GRRTM, PGGRRTT) and certain percentages (UAGRNT, RMINSTP) since they appear relatively normally distributed or well-spread

The transformed numerical variables pair plots look much better than that of the original numerical variables in the following pages. The plots will not be included in the knitted version but the code is available in the Rmd.

3. Best Subset Selection

Group Variables into Categories

We have too many predictors to run one model. So we split into groups as following (Note that this grouping is strictly for best subset and grouping in the R Shiny App should follow the code book).

- Group 1: Enrollment & Demographics
- Group 2: Outcomes & Institutional Characteristics

- Group 3: Financial Aid & Grant
- Group 4: Institutional / Structural Indicators
- Group 5: Geography
- Group 6: Program Mix

```
## Manual groupings count matches nubmer of columns in public4: FALSE
```

Next, we run best subset on each group and find the best variables from each group:

```
## Warning in leaps.setup(x, y, wt = wt, nbest = nbest, nvmax = nvmax, force.in =
## force.in, : 2 linear dependencies found
```

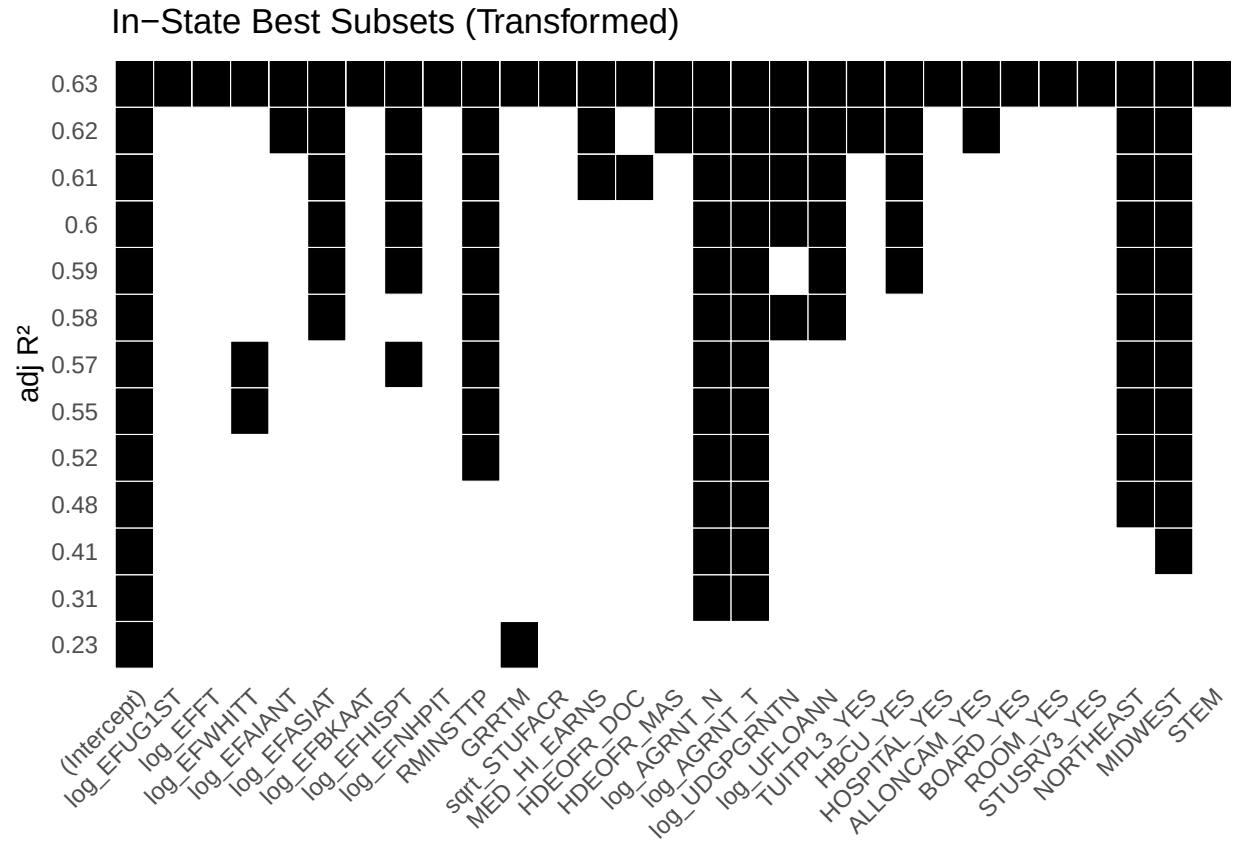
```
## Reordering variables and trying again:
```

We also run into a potential problem that is a variable could be not important in its grouping best subset but could have value when paired with others. Right now, our analysis does *not* take care of this issue and does *not* consider interactions. The most important variables are in the pool (lower bic, better model):

```
## [1] "log_EFUG1ST" "log_EFFT" "log_EFWHITT" "log_EFAIANT"
## [5] "log_EFASIAT" "log_EFBKAAT" "log_EFHISPT" "log_EFNHPIT"
## [9] "RMINSTTP" "GRRTM" "sqrt_STUFACR" "MED_HI_EARNS"
## [13] "HDEOFR_DOC" "HDEOFR_MAS" "log_AGRNT_N" "log_AGRNT_T"
## [17] "log_UDGPGRNTN" "log_UFLOANN" "TUITPL3_YES" "HBCU_YES"
## [21] "HOSPITAL_YES" "ALLONCAM_YES" "BOARD_YES" "ROOM_YES"
## [25] "STUSRV3_YES" "NORTHEAST" "MIDWEST" "STEM"
```

Final Best Subset

Now, we run another best subset with the variables we deem important from the above analysis.



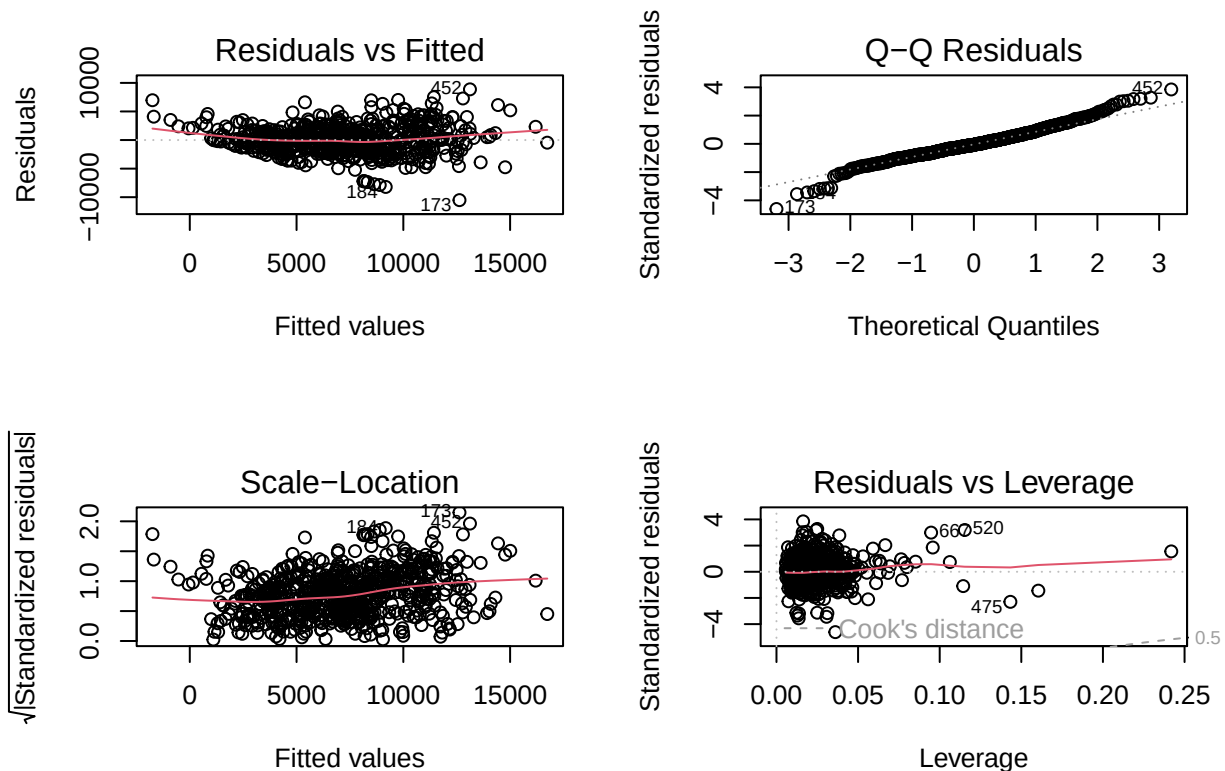
Linear Model

We now create a linear model with the selected variables from the best subset. We will include all n-1 levels of a categorical variable we transformed into dummies.

```
##
## Call:
## lm(formula = TUITION2 ~ log_EFUGCNT + log_EFASIAT + log_EFHISPT +
##      GRRTM + HDEOFR_DOC + RSCH_HIGH + RSCH_INSTS + log_AGRNT_N +
##      log_AGRNT_T + log_UDGPGRNTN + log_UFLOANN + HBCU_YES + RELAFFIL_NO +
##      NORTHEAST + MIDWEST + RMINSTTP + MED_HI_EARNS - UNITID, data = public4)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10513.6  -1453.2   -160.3   1291.9   8867.8
##
## Coefficients: (1 not defined because of singularities)
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -31461.008   3360.648  -9.362  < 2e-16 ***
## log_EFUGCNT    830.187   359.569   2.309   0.0212 *
## log_EFASIAT    464.239   108.232   4.289  2.04e-05 ***
## log_EFHISPT   -536.509   114.998  -4.665  3.69e-06 ***
## GRRTM         -13.646     7.920  -1.723   0.0853 .
## HDEOFR_DOC     458.634   297.047   1.544   0.1230
```

```
## RSCH_HIGH      695.574    391.061    1.779    0.0757 .
## RSCH_INSTS     448.427    299.281    1.498    0.1345
## log_AGRNT_N   -4198.267    386.856   -10.852   < 2e-16 ***
## log_AGRNT_T    4525.685    340.833    13.278   < 2e-16 ***
## log_UDGPRNTN  -1655.855    322.733    -5.131   3.74e-07 ***
## log_UFLOANN     464.745     94.077     4.940   9.78e-07 ***
## HBCU_YES      -2234.176    479.974    -4.655   3.88e-06 ***
## RELAFFIL_NO      NA         NA         NA         NA
## NORTHEAST      1964.802    258.767     7.593   1.00e-13 ***
## MIDWEST        2097.988    241.511     8.687   < 2e-16 ***
## RMINSTTP       -31.921      6.655    -4.797   1.97e-06 ***
## MED_HI_EARNS    956.182    305.182     3.133    0.0018 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2322 on 700 degrees of freedom
## Multiple R-squared:  0.6251, Adjusted R-squared:  0.6165
## F-statistic: 72.95 on 16 and 700 DF,  p-value: < 2.2e-16
```

We also generate plots to validate the model.



Summary:

$$R^2 = 0.62.$$

The model suggests that Public in-state tuition behaves differently than private tuition because it is heavily influenced by state policy and institutional mission:

- Research Focus: Similar to private model, RSCH_HIGH and RSCH_INSTS are both positive and significant ($p < 0.05$), confirming that research-intensive public universities charge more.
- Geographic Regions: NORTHEAST and MIDWEST have extremely high t-values (6.59 and 7.87), indicating that public tuition is much higher in these regions compared to the South and West.
- HBCU Status: Unlike in the private model, HBCU_YES is significant and negative, suggesting public HBCUs maintain significantly lower in-state tuition.
- Loans: Similar to private model, AGRNT_N and AGRNT_T are important.

Next step: justifying outliers, validating models, run this for out-of-state tuition